

Yequan Zhao

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Education

University of California Santa Barbara, CA, USA	<i>Sept. 2023-Now</i>
Ph.D. in Electrical & Computer Engineering	GPA: 3.96/4.0
Huazhong University of Science and Technology, Wuhan, China	<i>Sept. 2019-Jun. 2023</i>
B.Eng. in Integrated Circuits Design and Integrated Systems	GPA: 3.94/4.0

Research Experiences

Professor Zheng Zhang's Group, University of California, Santa Barbara	
<i>Graduate Research Assistant</i>	<i>Sept. 2023-Present</i>

Memory-efficient On-Device Training via Backpropagation-free Approach

Training as a plug-in over inference accelerator: fully-quantized, no additional memory

- Full model training on microcontrollers (MCUs) with tiny memory budget (256-KB SRAM). [TODAES'25](#)
- Full model fine-tuning on truly quantized LLM [EMNLP'25](#)
- Real-time PDE solver on optical computing hardware [TODAES'26](#)

Efficient LLM Training:

- Spectral preconditioned update: LoRA budget, full finetune accuracy, even faster converge [NeurIPS'26](#)
- Efficient optimizers: Boosted Muon [EMNLP'26 \(Oral\)](#)

Professional Experiences

Amazon Lab 126	<i>Jun. 2026 – Sept. 2026</i>
<i>Applied Scientist Intern</i>	
Supervisor: Sankalp Dayal & Tong Qin	
Project: Efficient mixture-of-experts (MoE) model serving on edge GPU.	
Amazon Lab 126	<i>Jun. 2025 – Sept. 2025</i>
<i>Applied Scientist Intern</i>	
Supervisor: Sankalp Dayal & Tong Qin	
Project: Large language model compression via tensor compression and knowledge distillation.	
Hewlett Packard Enterprise	<i>Jul. 2024 – Sept. 2024</i>
<i>Research Associate Intern</i>	
Supervisor: Dr. Xian Xiao & Dr. Geza Kurczveil (HPE Large-scale Integrated Photonics Lab)	
Project: Experimental Demonstration of an Optical Neural PDE Solver via On-Chip PINN Training	

Publications

[NeurIPS'26](#) Yequan Zhao, Ruijie Zhang, Liyan Tan, Niall Moran, Tong Qin, Zheng Zhang. FuRA: Full-Rank Parameter-Efficient Fine-Tuning with Spectral Preconditioning. *Advances in Neural Information Processing Systems* 38 (2026) (Just Accepted)

[EMNLP'26 \(Oral\)](#) Liu, Ziyue, Ruijie Zhang, Zhengyang Wang, **Yequan Zhao**, Yupeng Su, Zi Yang, and Zheng Zhang. "Muon²: Boosting Muon via Adaptive Second-Moment Preconditioning." In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing* (Just Accepted)

[TODAES'26](#): **Yequan Zhao**, Xinling Yu, Xian Xiao, Zhixiong Chen, Ziyue Liu, Geza Kurczveil, Raymond Beausoleil, Sijia Liu, and Zheng Zhang. Scalable Back-Propagation-Free Training of Optical Physics-Informed Neural Networks. *ACM Transactions on Design Automation of Electronic Systems* (TODAES).

MLSys'26: Tian, Jiayi, Seyedarmin Azizi, **Yequan Zhao**, Erfan Baghaei Potraghloo, Sean McPherson, Sharath Nittur Sridhar, Zhengyang Wang, Zheng Zhang, Massoud Pedram, and Souvik Kundu. "SkipKV: Selective Skipping of KV Generation and Storage for Efficient Inference with Large Reasoning Models." *Proceedings of Machine Learning and Systems* 8 (2026): 1496-1514.

TODAES'25: **Yequan Zhao**, Hai Li, Ian Young, and Zheng Zhang. "Poor Man's Training on MCUs: A Memory-Efficient Quantized Back-Propagation-Free Approach", *ACM Transactions on Design Automation of Electronic Systems (TODAES)*.

EMNLP'25: Jiajun Zhou, Yifan Yang, Kai Zhen, Ziyue Liu, **Yequan Zhao**, Ershad Banijamali, Ngai Wong, and Zheng Zhang. "QuZO: Quantized Zeroth-Order Fine-Tuning for Large Language Models". In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 5341-5359. 2025.

OFC'25: **Yequan Zhao**, Xian Xiao, Antoine Descos, Yuan Yuan, Xinling Yu, Geza Kurczveil, Marco Fiorentino, Zheng Zhang, and Raymond G. Beausoleil. "Experimental Demonstration of an Optical Neural PDE Solver via On-Chip PINN Training", *2025 Optical Fiber Communications Conference and Exhibition (OFC)*

APL Machine Learning: Xian Xiao*, **Yequan Zhao***, Yuan Yuan, Geza Kurczveil, Marco Fiorentino, Raymond G. Beausoleil, and Zheng Zhang. "TOMFuN: A Tensorized Optical Multimodal Fusion Network"

TCAS-I: Bingqiang Liu, Jipeng Wang, **Yequan Zhao**, et. al. "An Energy-efficient, High-frame-rate and Reconfigurable EKF-SLAM Processor with Full Acceleration Supporting Large-Scale Mapping for Autonomous Mobile Robots".

TMLR: Xinling Yu, Sean Hooten, Ziyue Liu, **Yequan Zhao**, Marco Fiorentino, Thomas Van Vaerenbergh, Zheng Zhang. "Separable Operator Networks".

CLEO'23: **Yequan Zhao**, Xian Xiao, Geza Kurczveil, Raymond G. Beausoleil, and Zheng Zhang. "Tensorized Optical Multimodal Fusion Network", *CLEO: Science and Innovations*. Optica Publishing Group, 2023.

Awards

ACM TODAES Rookie Author of the Year	2026
UCSB ECE Wallace Kou Graduate Fellowship	2026

Services

Reviewer for NeurIPS 26', AAAI 26', IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) IEEE Transactions on Artificial Intelligence (TAI)